

White paper: is Architecture C biologically motivated? A latent-class assessment, with application to the prazosin/PTSD setting

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pmsimstats team. A cross-paper note relating the combined DGP architecture of paper 01 to the latent-class formulation of paper 03, and assessing which specification best fits the prazosin/PTSD reference application. Written to be readable without prior exposure to either paper; terms are defined at first use and collected in a glossary at the end.

1. Purpose and scope

Paper 01 defines a third data-generating architecture, Architecture C, which switches on a mean-moderation channel (parameter $c_{bm,A}$) and a differential-correlation channel (parameter $c_{bm,B}$) at the same time. Its stated justification is mechanistic: the two channels are ‘mechanistically distinct and need not scale in proportion’, so a drug could activate one, both, or neither.

Paper 03 develops a latent-class account of the same phenomenon. This note asks what paper 03’s theory implies for paper 01’s Architecture C. The conclusion qualifies paper 01’s justification: combining the two channels is defensible, but not for the reason paper 01 gives, and not in the form paper 01 implements.

Two results in this note are new and were computed for it. Section 5 works out, by simulation, which combinations of the two parameters a two-class biological mechanism can actually produce, and finds that the answer is a restricted wedge that excludes pure Architecture B entirely. Section 9.1 resolves a calibration question about the existing code and finds that paper 01’s two architectures are not matched on effect size as closely as the manuscript implies.

A note on terminology. A *data-generating process* (DGP) is the recipe the simulation uses to manufacture fake trial data. The *analysis model* is the statistical model we then fit to that fake data, exactly as we would fit it to a real trial. Power is estimated by generating many trials, fitting the analysis model to each, and counting how often the test comes out significant. The subject of this note is the recipe, not the analysis.

2. What we are estimating, and the three things that can change it

Everything here follows from one question: what is the trial actually testing?

The analysis model is fixed across all of paper 01:

$$Sx \sim bm + t + Dbc + bm:Dbc$$

where Sx is the symptom score, bm is the patient’s baseline biomarker, t is time, and Dbc is drug exposure, equal to 1 while the patient is on drug and decaying smoothly toward 0 after they stop. The term of interest is $bm:Dbc$, the interaction. Its coefficient answers the clinical question: **does the drug work better in patients with a higher biomarker?** That coefficient is the *estimand*, the quantity we are trying to estimate, and the trial succeeds when we can distinguish it from zero.

Power for that test is governed, to a good approximation, by

$$\text{power} \approx f\left(\frac{\text{size of the true coefficient}}{\text{standard error of its estimate}}\right)$$

so anything the recipe does must show up in one of three places:

1. **Signal: the size of the coefficient.** How much does a one-unit increase in the biomarker actually change the drug’s effect on the outcome? This is the numerator.
2. **Noise: the scatter around it.** How variable are patients’ responses once you know their biomarker and drug status? More scatter means a larger standard error, which is the denominator.
3. **Information: how much the off-drug timepoints contribute.** The interaction is identified by contrasting on-drug against off-drug observations. If carryover blurs that contrast, or if the biomarker stops predicting anything once the drug is withdrawn, the off-drug observations stop pulling their weight.

These three buckets organize the whole note. Two recipes that agree in all three give the same power; two that differ in any one may not.

2.1 The quantities that populate the three buckets

Conditional mean. The average outcome among patients who share a given biomarker value, written $E[BR | B = b]$. Read it as: ‘among patients whose biomarker equals b , what is the mean biological response?’ The word ‘conditional’ means only ‘restricting attention to patients with that biomarker value’.

Conditional mean gradient. How fast that average changes as the biomarker increases. If patients with a biomarker of 2 respond one unit better than patients with a biomarker of 1, the gradient is 1 per biomarker unit. *This is why we care:* the gradient is essentially what the `bm:Dbc` coefficient estimates. It is bucket 1, the signal. Two recipes with the same gradient feed the same signal into the test, whatever else differs between them.

Conditional variance. How spread out responses are among patients who share a biomarker value, written $\text{Var}(BR | B = b)$. *This is why we care:* it is the residual scatter the model must see through, so it drives the standard error. It is bucket 2, the noise.

Buckets 1 and 2 are what statisticians call the *first moment* (the mean) and the *second moment* (the variance and covariance) of a distribution. When paper 01 says Architecture A puts the signal ‘in the first moment’ and Architecture B puts it ‘in the second moment’, this is the distinction being drawn. Section 3 shows that the distinction, stated that way, is not accurate.

Carryover and washout. *Washout* is the period after a patient stops the drug. *Carryover* is residual drug effect persisting into it. The *carryover half-life* is how long that residual takes to fall by half. Bucket 3 concerns what happens to the biomarker’s predictive value during washout.

3. What Architectures A and B actually differ in

The usual summary is that Architecture A encodes the biomarker effect in the mean and Architecture B in the variance. That is not what the two recipes do.

On drug, the two produce the same conditional mean gradient. Under Architecture A the recipe draws the response independently of the biomarker and then adds a shift of $c_{bm} z_B \sigma_{BR}$, where z_B is the patient’s biomarker in standard-deviation units and σ_{BR} is the response standard deviation. Under Architecture B the recipe instead builds a correlation between biomarker and response into the joint draw, giving a conditional mean of $\mu_{BR} + c_{bm}(\sigma_{BR}/\sigma_B)(b - \mu_B)$. Substituting $z_B = (b - \mu_B)/\sigma_B$ shows these are the same quantity. The calibration comment at `R/generateData.R:129-131` states the correspondence explicitly, and it is the reason the two architectures are comparable at matched c_{bm} at all.

So both are identical in bucket 1. They differ in buckets 2 and 3:

- **Bucket 2, the noise.** Under Architecture B, knowing the biomarker genuinely tells you something about the response, which narrows the range of plausible responses: $\text{Var}(BR | B) = \sigma_{BR}^2(1 - c_{bm}^2)$, shrunk. Under Architecture A the biomarker and response are drawn independently and the effect is bolted on as a fixed shift, sliding each patient’s response up or down without making them any more predictable: $\text{Var}(BR | B) = \sigma_{BR}^2$, unchanged.
- **Bucket 3, the off-drug information.** Architecture A’s shift is controlled by a hard on/off switch, `onDrug := (tod > 0)` (`R/generateData.R:141, :166`), so it is exactly zero at every off-drug timepoint. Architecture B’s correlation fades gradually, as $c_{bm} e^{-\lambda t_{sd}}$, where t_{sd} is time since discontinuation.

The accurate description is therefore: **Architectures A and B are two different noise-and-washout completions of one shared signal.** This reframing decides the biological question, because it means Architecture C’s two parameters are not independent knobs. They both turn the same signal dial.

4. What a latent-class mechanism produces

4.1 The vocabulary

A *latent class* is an unobserved subgroup. ‘Class’ means a discrete category, such as responder versus non-responder; ‘latent’ means we never observe which category a patient is in. It is the formal version of the clinical intuition that a drug works in some patients and not others.

A *gating function*, written $\pi(B)$, is the probability that a patient with biomarker value B belongs to the responder class. It ‘gates’ the drug effect in the sense of a valve: it controls how likely the effect is to be switched on for that patient. Its shape is the whole story:

- A *flat* gating function means the biomarker carries no information about who responds.

- A *step* gating function means the biomarker is a perfect classifier with a threshold: everyone above a cutoff responds, nobody below.
- A *graded* function sits in between: higher biomarker values make response more likely without guaranteeing it.

The *gating slope* measures how sharply the function rises, that is, how close it is to a step. This single quantity turns out to determine which architecture is the better approximation.

4.2 The three signatures

Suppose the truth is: each patient belongs to a hidden class $C_i \in \{0, 1\}$; the chance of being a responder is $\pi(B_i)$; and on drug the two classes have mean responses μ_0 and μ_1 , separated by $\Delta = \mu_1 - \mu_0$.

We never observe the class, so what the trial sees is the average over both possibilities, weighted by how likely each is. That produces one signature in each bucket:

1. **A mean gradient (bucket 1).** The observed average response at biomarker value b is $\mu_0 + \Delta \pi(b)$: the non-responder mean, plus the extra benefit, weighted by the chance of being a responder. Because π rises with b , so does average response. The steepness of that rise is roughly Δ times the gating slope, so a bigger class difference and a sharper gate both strengthen the signal.
2. **Variance that depends on the biomarker (bucket 2).** The spread of responses is $\Delta^2 \pi(b)(1 - \pi(b))$ plus ordinary within-class variability. The extra term is largest where $\pi(b) \approx 0.5$. The intuition is direct: a patient whose biomarker leaves them on a coin flip between the two classes is the least predictable, while patients at either extreme are nearly certain to be in one class and vary less. Variance that changes with a predictor this way is called *heteroscedasticity*.
3. **A class label that persists (bucket 3).** A patient's class is a property of the patient, not of their current drug status. It does not wash out. Paper 03 identifies this as the reason some information about who responds survives discontinuation and stays recoverable from the patient's full trajectory.

Both architectures capture signature 1. **Neither captures signature 2:** Architecture B shrinks variance by the same factor for every patient, Architecture A does not shrink it at all, and the truth shrinks it by an amount depending on where the patient sits on the gating curve. **Neither captures signature 3:** both are built timepoint by timepoint from correlations and shifts, with no persistent label attached to the patient.

4.3 Summary comparison

Recipe	Bucket 1: signal	Bucket 2: noise	Bucket 3: off drug
Architecture A	gradient $c z_B \sigma_{BR}$	unchanged	exactly zero
Architecture B	same gradient	shrunk by $1 - c^2$	fades as $e^{-\lambda t_{sd}}$
Architecture C	sum of both gradients	partial shrinkage	mixed
Two-class mixture	gradient $\Delta \pi'(b)$	varies with b	label persists

5. Worked example: which parameter pairs can a mixture actually produce?

Section 3 argued that Architecture C's two parameters are coupled rather than independent. That argument can be checked numerically.

The experiment is simple. Simulate a genuine two-class mixture with a known gating slope, then ask what parameter value each architecture would need in order to reproduce what the mixture produced. For Architecture A the relevant summary is the regression slope of response on standardized biomarker, divided by the within-class response SD. For Architecture B it is the observable correlation between biomarker and response. Call these c_A and c_B . Two million patients per cell were simulated, so the values below are precise to about three decimal places.

Table 1. Gating slope against the implied parameter pair, at class separation $\Delta = 1$ within-class SD, with half of patients in each class. The last two columns give the conditional SD of response at the centre of the biomarker distribution and two SDs out.

Gating slope	c_A	c_B	c_B/c_A	SD at $b = 0$	SD at $b = 2$
0.25	0.061	0.055	0.895	1.119	1.107
0.5	0.118	0.106	0.895	1.120	1.091
1.0	0.206	0.184	0.895	1.119	1.049
2.0	0.302	0.270	0.896	1.117	1.012
4.0	0.366	0.327	0.895	1.116	0.995
8.0	0.390	0.349	0.894	1.117	0.998

Three things are visible.

First, the two parameters rise together and their ratio never moves. Sharpening the gate from nearly flat to nearly a step raises c_A six-fold and c_B six-fold, but the ratio stays pinned at 0.895. The mixture traces a straight ray out from the origin. It never travels along either axis.

Second, heteroscedasticity appears exactly as predicted, and it is modest. At a nearly flat gate the conditional SD is the same at the centre and in the tail. As the gate sharpens, the tail SD falls to about 1.00 while the centre stays at about 1.12, so patients near the 50/50 point end up roughly 12 percent more variable than patients at the extremes. That is a real departure from both architectures, both of which assume constant conditional variance, but at this class separation it is not a dramatic one. This is a point in favor of the correlation approximation in the graded regime, and it is consistent with paper 03’s claim that a single multivariate normal reproduces a mixture to second order.

Third, paper 03’s bound is exact. Paper 03 proves that the achievable correlation satisfies $c_{bm}^2 = f_B \times f_{BR}$, the product of the between-class variance fractions in biomarker and in response. Computing that bound for each row reproduces the simulated c_B to three decimals in every case (0.055, 0.106, 0.185, 0.271, 0.326, 0.348 against the observed values above). The proposition holds numerically.

5.1 What sets the direction of the ray

Table 1 fixes the class separation. Varying it shows what controls the ray’s angle.

Table 2. Implied parameter pairs at two gating slopes, across class separations.

Δ	c_A (slope 1)	c_B (slope 1)	c_A (slope 8)	c_B (slope 8)	ratio
0.5	0.103	0.100	0.195	0.190	0.970
1.0	0.206	0.185	0.390	0.348	0.895
1.5	0.309	0.247	0.583	0.467	0.801
2.0	0.412	0.292	0.778	0.550	0.708
3.0	0.620	0.345	1.167	0.648	0.555

The ratio depends only on the class separation, and is identical at both gating slopes. So the structure is clean: **class separation sets the direction of the ray, gating slope sets how far along it you travel.** The closed form behind the ratio column is

$$\frac{c_B}{c_A} = \frac{\sigma_w}{\sqrt{\sigma_w^2 + \Delta^2 p(1-p)}}$$

where σ_w is within-class SD and p the responder fraction. The ratio is 1 when there is no class separation and falls toward 0 as separation grows.

5.2 The consequence for Architecture C’s grid

This ratio is always at most 1, so every reachable point satisfies $c_B \leq c_A$. The set of parameter pairs a two-class mechanism can produce is the **wedge between the 45-degree line and the c_A axis**. Within that wedge:

- **Pure Architecture B is unreachable.** The c_B axis, where the mean channel is zero and only the correlation channel is active, corresponds to no two-class mixture at all. Producing correlation without a mean gradient would require the gating function to be uncorrelated with the biomarker, which defeats the purpose of having a predictive biomarker.

- **Pure Architecture A is the limiting case** of very large class separation, where the mean gradient dominates and the correlation contribution becomes negligible by comparison.
- **The mean channel always leads.** For any two-class mixture, $c_A > c_B$. A mixture is therefore always closer to Architecture A than to Architecture B in these coordinates.

Table 2 also puts paper 01’s headline parameter value in context. Reaching $c_B = 0.45$ requires a nearly step-like gate together with a class separation of about 1.5 within-class SDs (giving $c_B = 0.467$). At a moderate gate, even a separation of 3 SDs only reaches $c_B = 0.345$. So $c_{bm} = 0.45$ is a demanding assumption: it needs both an unusually clean classifier and a large responder advantage. And the mixture that delivers $c_B = 0.467$ delivers $c_A = 0.583$ at the same time. The pure-B cell that carries paper 01’s most quoted result corresponds to no two-class biology whatever.

6. Assessment: is combining the two channels biologically motivated?

6.1 The unconstrained grid is not produced by any single mechanism

Since both parameters feed the same conditional mean gradient (Section 3), turning both up adds two signals together. Paper 01 observes the consequence without drawing the inference: at $(c_{bm,A}, c_{bm,B}) = (0.45, 0.45)$, ‘the total interaction effect on the BR mean is approximately double that of either pure architecture’.

Across most of the 3×3 grid, moving toward the interior is therefore mostly **making the effect bigger**, with noise and washout structure changing alongside. It is not switching on a second biological channel while holding the first fixed. Section 5 sharpens this: the reachable set is a wedge, so the grid’s corner cells and much of its interior correspond to no two-class mechanism at all. Paper 01’s stated reason for preferring the unconstrained grid, that it ‘spans the full $(c_{bm,A}, c_{bm,B})$ space’, is the wrong criterion if much of that space corresponds to no biology.

6.2 The mixing-weight version is coherent

Paper 01 defines, then declines to use, a constrained alternative: set $c_{bm,A} = (1 - \alpha)c$ and $c_{bm,B} = \alpha c$, where c is a fixed total effect size and α runs from 0 to 1. At $\alpha = 0$ this is pure Architecture A, at $\alpha = 1$ pure Architecture B, and in between a genuine blend.

This version holds bucket 1 fixed and varies only buckets 2 and 3, which is exactly what separates the two architectures, so sweeping α answers a clean question rather than a confounded one. It is also the sweep paper 03’s theory calls for, since paper 03 shows that a graded gate is well approximated by the correlation recipe whereas a near-step gate is better approximated by a thresholded mean-moderation recipe (**03-latent-class-mixture-application/report.Rmd:1202**). The weight α can thus be read as a stand-in for gating sharpness, a property of the biology one could investigate, rather than as a mixture of two separate mechanisms.

One caveat follows from Section 5.2. Because reachable pairs satisfy $c_B \leq c_A$, the biologically supported part of the mixing line is its lower half, $\alpha \leq 0.5$. Values above that correspond to correlation exceeding mean gradient, which no two-class mechanism produces.

6.3 What no Gaussian recipe captures

Neither pure architecture, and so no point on the mixing line, reproduces the biomarker-dependent variance or the persistent class label of Section 4.2. Architecture C should accordingly not be described as covering the latent-class possibility. If the worry is that the biomarker tags a responder subtype, the right response is to simulate the subtype structure directly. Paper 03’s heterogeneous-random-slopes form is the appropriate target, since it admits latent-class structure in the recipe while leaving the analysis model an ordinary `nlme::lme` fit.

7. When does the correlation approximation fail?

Paper 03 names three regimes in which a single multivariate normal cannot stand in for a mixture. Each has a distinct empirical signature, which makes the taxonomy usable rather than merely cautionary.

Bimodal responses. When the two classes are well separated, so non-responders show near-zero effect and responders a large one, the distribution of on-drug response has two peaks rather than one. No single normal distribution matches the tails of a two-peaked distribution. Paper 03 notes that in this regime power under the true mixture *exceeds* the normal approximation’s prediction, because a correctly specified mixture analysis can

exploit the two peaks. The clinical analogue is a subpopulation that genuinely does not respond, as in some targeted oncology indications.

A strong class-membership gradient. When the gate approaches a step, the biomarker behaves almost like a class label. A thresholded mean-moderation specification then becomes the better approximation, because the relationship is dominated by its mean component. The clinical analogue is a biomarker with near-perfect sensitivity and specificity for the responsive phenotype. Section 5 supports this from the other direction: as the gate sharpens, both implied parameters saturate, and the mean channel remains the larger throughout.

Class-varying covariance. If autocorrelation, residual variance, or placebo-response parameters differ between classes, no single-component normal can represent the structure, and an analysis assuming constant covariance is misspecified regardless of where it puts the biomarker interaction. This is the latent-class analogue of heteroscedastic residual variance, ordinarily handled with `varIdent` or `weights` in mixed-model practice.

Only the second of these regimes is addressed by moving along the mixing line. The first and third require the mixture recipe itself.

8. Assessment for the prazosin/PTSD setting

Paper 01 concludes that the prazosin application ‘is an Architecture B story’ because resting systolic blood pressure (SBP) is ‘a proxy for an underlying neurobiological state, not a direct determinant of drug pharmacokinetics’. Four considerations qualify that conclusion.

8.1 The classification is drawn on the wrong axis

Pharmacokinetics (PK) is what the body does to the drug: absorption, distribution, clearance, and hence the concentration achieved. *Pharmacodynamics* (PD) is what the drug does to the body: how much effect a given concentration produces, which depends on how much target is available to act on.

Paper 01’s argument rules out Architecture A on PK grounds. But its own Architecture A criteria are not confined to PK: the worked examples include receptor density, ‘if a biomarker measures how many target receptors a patient has, more receptors means proportionally more drug effect’, which is a PD quantity. Noradrenergic tone is a PD substrate of exactly this kind. Prazosin blocks alpha-1 adrenergic receptors, so a patient with more alpha-1 signaling to block plausibly gets a proportionally larger effect. On paper 01’s own taxonomy, that is an Architecture A mechanism.

The section is internally inconsistent here. It lists ‘alpha-1 receptor distribution’ among the sources of slippage that supposedly make SBP a mere proxy, but alpha-1 receptor density is the paper’s own canonical Architecture A example. The sound argument for Architecture B in this setting is not that SBP fails to determine drug levels; it is that SBP measured in the arm is a noisy readout of noradrenergic activity in the brain, and prazosin’s benefit in PTSD is thought to be centrally mediated. That is an argument about measurement slippage, and it coexists with a real PD component rather than replacing it. Both being true argues for an intermediate mixing weight, not for pure Architecture B. Section 5.2 reinforces this from theory: pure Architecture B is not something a responder-subtype mechanism can produce in the first place.

8.2 The clinical evidence points past both architectures to a mixture

The evidence base paper 01 cites is itself the signature of a responder subgroup diluted by non-responders. The earlier single-site trials reported benefit; the multisite PACT trial did not replicate, and paper 01 notes this ‘has been interpreted as consistent with effect heterogeneity, namely that prazosin works in some patients but not others’. That is a latent-class hypothesis stated in plain language.

This points the modeling error in an unexpected direction. Under paper 03’s bimodal-response regime (Section 7), power under the true mixture exceeds what the normal approximation predicts. If the prazosin responder subgroup is real and reasonably distinct, Architecture B may be **understating** achievable power, the opposite of the direction paper 01’s carryover warning points.

This is checkable cheaply. Plotting the distribution of on-drug response in the existing Raskind datasets would distinguish a single broad peak, which supports a correlation-leaning specification, from two separate peaks, which supports neither pure architecture and calls for the mixture recipe. Section 5 gives the quantitative version of the same check: an estimate of the responder advantage in within-class SD units and of the responder fraction is enough to locate the application on the wedge and read off the implied parameter pair.

8.3 Carryover here is behavioral, and neither washout model fits

Prazosin is cleared from the blood in a few hours, yet the simulations use carryover half-lives of 0.5 to 1.0 weeks. Whatever persists across those weeks is not the drug. It is the behavioral and physiological state the drug produced: sleep architecture and nightmare frequency re-equilibrating after discontinuation.

That undermines both washout models, in opposite ways. Architecture B fades the biomarker-response *association* exponentially, which is right when the drug effect is gone and the association goes with it. But under behavioral carryover the benefit itself persists while the drug is absent, so a responder keeps responding for some weeks and the association should persist too, plausibly at near full strength through the first off-drug week. Architecture B's fade is thus too fast here. Architecture A is worse: its hard on/off switch sets the biomarker-scaled benefit to exactly zero at every off-drug timepoint while the average benefit lingers, which is not a coherent account of behavioral carryover under any reading.

The specification that would fit is a mean-moderation channel fading with the same exposure profile the analysis model already uses for `Dbc`, giving $\Delta BR_{it} = c_{bm,A} z_{B,i} \sigma_{BR} e^{-\lambda t_{sd}}$ off drug instead of zero. **This variant is not implemented.** The loop at `R/generateData.R:165-170` applies the shift only where `onDrug` is true, and neither the constant nor the trajectory setting of `moderation_scaling` changes that gate.

The consequence deserves emphasis. Because Architecture A's signal does not decay at all in the recipe, the 1 to 3 percent power losses paper 01 reports for Architecture A come entirely from the analysis model's exposure variable shrinking, not from any erosion of the generated signal. That is a coding choice being reported as a finding about mean-moderation biology, and it should be labeled as such.

8.4 Recommended specification

Three independent considerations point the same way, and all three suggest the pure-Architecture-B OL+BDC figure of 0.539, a 30.6 percent loss, is a pessimistic bound rather than a planning estimate:

1. SBP plausibly carries a real pharmacodynamic component alongside its proxy role (Section 8.1), so the mean channel is not empty, and Section 5.2 shows it is always the larger channel under a responder-subtype mechanism.
2. Behavioral carryover implies the association persists longer than Architecture B allows (Section 8.3).
3. If the responder subgroup is real and distinct, the true mixture supports higher power than the correlation approximation predicts (Section 8.2).

Sizing a trial on 0.539 will over-power it. Sizing on the pure Architecture A figure of 0.730 will under-power it if the gate is smooth and arm-to-brain slippage is large. The defensible procedure is to report power along the constrained mixing line at fixed total effect size, restrict attention to $\alpha \leq 0.5$ per Section 6.2, treat the resulting interval as the planning range, and adopt an intermediate weight as the working estimate pending evidence on gating sharpness.

8.5 The design decision does not have to wait on any of this

The choice of trial design is robust to the architecture question even though the power estimate is not.

In paper 01's Architecture C grid, at full signal and one-week carryover, the Hybrid design attains 0.870 against 0.688 for OL+BDC and 0.528 for the crossover. In the Section 3.1 tables the Hybrid design is likewise first under both pure architectures at every carryover level. Hybrid wins under every specification examined.

The second-versus-third ordering is *not* robust: under Architecture A at one-week carryover, OL+BDC (0.730) marginally exceeds the crossover (0.725), whereas under Architecture B the crossover (0.723) exceeds OL+BDC (0.539) by a wide margin. The architecture question governs the power estimate and the runner-up ranking, but not the primary recommendation. For the prazosin application the recommendation is Hybrid, and it can be made now.

9. How to tell which architecture is right

9.1 A calibration check on the existing code

An open question from an earlier draft has now been resolved by direct simulation against the package. Architecture A adds its shift on top of a draw that already has full variance, so it should inflate total response variance

by a factor of $(1 + c_{bm}^2)$, whereas Architecture B builds the association into the draw and leaves total variance alone. If so, the two architectures are not matched on observable effect size at nominal parity.

Running `generateData` at $N = 200,000$ on an eight-timepoint open-label design confirms it.

Table 3. Observable correlation and response SD at an on-drug timepoint, by architecture and nominal parameter value.

Architecture	nominal c_{bm}	induced correlation	SD of response
B (MVN)	0.45	0.446	4.99
A (mean moderation)	0.45	0.405	5.47
B (MVN)	0.30	0.294	4.99
A (mean moderation)	0.30	0.281	5.21

Architecture B lands on its nominal value. Architecture A at nominal 0.45 delivers an observable correlation of 0.405, about 9 percent low, and inflates the response SD from 4.99 to 5.47. **Paper 01’s head-to-head comparison is therefore not exactly effect-size matched, and Architecture A is the handicapped arm in every cell**, carrying both a smaller induced association and more response variance.

The direction of the headline conclusion survives, since Architecture A is disadvantaged and still shows the greater carryover robustness. But two things follow. The baseline power gap at no carryover (0.751 for A against 0.777 for B at OL+BDC) is at least partly this artifact rather than a property of the architectures. And because power loss is not scale-free, comparing losses measured from different baselines contaminates the difference-of-differences contrast to some degree. The comparison should be re-run with A’s parameter adjusted so the two induce matching observable correlations.

9.2 Diagnostics available from real trial data

Three checks discriminate among the candidates without needing a new trial:

1. **Shape of the on-drug response distribution.** Two peaks indicate a separated mixture, for which neither architecture is adequate. One broad peak is consistent with graded gating.
2. **Conditional variance across biomarker strata.** Split patients into biomarker tertiles and compare response SD. A mixture predicts the middle stratum is the most variable (Section 5, Table 1). Both architectures predict no difference. The effect is modest, roughly 12 percent at a class separation of 1 SD, so this check needs a reasonable sample.
3. **Does the biomarker still predict response during washout?** If the association survives discontinuation at close to full strength, the persistent-label picture is right and both architectures’ washout models are wrong. This is the most decisive check and the most directly relevant to the prazosin setting.

None of these has been run here.

10. Recommendations

1. Keep Architecture C, but move the production grid onto the constrained mixing line at fixed total effect size, restricted to $\alpha \leq 0.5$. Reserve the unconstrained grid for the two boundary cells, which are useful as implementation checks because they reproduce the pure architectures exactly.
2. Re-run the Section 3.1 head-to-head with Architecture A’s parameter adjusted for the calibration gap in Table 3, so the two architectures induce matching observable correlations.
3. Implement an exposure-decayed version of the mean-moderation channel and re-run Architecture A under it. Until then, describe the 1 to 3 percent Architecture A losses as conditional on a hard on/off gate.
4. State explicitly that Architecture C does not span the latent-class possibility, and cross-reference paper 03’s heterogeneous-random-slopes recipe as the construct that does.
5. Revise paper 01 Section 4.3 to argue the prazosin classification on the central-versus-peripheral measurement axis rather than the pharmacokinetic axis, and to acknowledge the pharmacodynamic component.
6. Reconcile paper 01 and paper 03 on carryover before either is submitted. Paper 03 warns that a covariance-only recipe overstates carryover-induced power loss; paper 01’s headline result is that loss. The two are reconcilable, since paper 01 conditions on the biomarker genuinely behaving like Architecture B, but the reconciliation should be written down rather than left to the reader.

11. Epistemic status

Verified by direct execution. The calibration results in Table 3 were produced by calling `generateData` from the installed package. The mixture results in Tables 1 and 2 were produced by simulation at two million patients per cell. Paper 03’s variance-fraction bound was checked numerically against Table 1 and reproduces the simulated correlation to three decimals.

Verified by inspection of source. The shared on-drug conditional mean gradient, the binary `onDrug` gate, and the absence of an exposure-decayed mean channel were confirmed against `R/generateData.R` at the line references given.

Analytic but not independently tested. The closed form for the ratio c_B/c_A in Section 5.1 is derived, and matches the simulated ratio column, but has not been checked against a separate derivation. The three signatures in Section 4.2 are standard results for finite mixtures.

Argued, not tested. The prazosin assessment in Section 8 argues from mechanism and from the cited clinical literature. No re-analysis of the Raskind trials was performed, and none of the three diagnostics in Section 9.2 has been run. All power figures quoted from paper 01 are read as published and were not recomputed.

Scope limits. The mixture simulations use a two-class logistic gate, a normally distributed biomarker, equal class sizes, and constant within-class variance. Section 7’s third regime, class-varying covariance, is therefore outside the scope of Tables 1 and 2, and the wedge result of Section 5.2 should not be assumed to hold when class sizes are strongly unequal. The gating intercept was held at zero throughout, which is what pins the responder fraction at one half; an offset gate would move the ratio.

12. Glossary

Carryover. Residual drug effect persisting after discontinuation.

Conditional mean. Average outcome among patients sharing a given biomarker value.

Conditional variance. Spread of outcomes among patients sharing a given biomarker value.

Data-generating process (DGP). The recipe used to manufacture simulated trial data.

Estimand. The quantity a study is trying to estimate; here, the `bm:Dbc` interaction coefficient.

Gating function. The probability that a patient with a given biomarker value belongs to the responder class.

Gating slope. How sharply the gating function rises; a measure of how nearly the biomarker acts as a perfect classifier.

Heteroscedasticity. Variance that changes with the value of a predictor.

Latent class. An unobserved patient subgroup, such as responders versus non-responders.

Pharmacodynamics (PD). What the drug does to the body; how much effect a given concentration produces.

Pharmacokinetics (PK). What the body does to the drug; absorption, distribution, and clearance.

Washout. The period after a patient stops the drug.

Appendix: reproducing Tables 1 and 2

```
set.seed(20260729)
n <- 2e6; sigma_w <- 1.0
b <- rnorm(n)

row_for <- function(delta, s) {
  cls <- rbinom(n, 1, plogis(s * b))
  br <- delta * cls + rnorm(n, 0, sigma_w)
  slope <- coef(lm(br ~ b))[2]
  c(delta = delta, s = s,
    cA = slope / sigma_w,
    cB = cor(b, br),
    ratio = cor(b, br) / (slope / sigma_w))
}
```

```
grid <- expand.grid(delta = c(0.5, 1.0, 1.5, 2.0, 3.0),  
  s = c(1.0, 8.0))  
round(t(mapply(row_for, grid$delta, grid$s)), 3)
```

Rendered on 2026-07-29 at 17:18 PDT. Source: ~/prj/res/36-pmsimstats-ng/pmsimstats-ng/analysis/report/01-dgp-mean-moderation-vs-mvn/whitepaper-architecture-c-latent-class-assessment.md